

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [2]: df = pd.read_csv("Advertising.csv")
df.head()
df.iloc[:5, [1, -1]]
```

```
Out[2]:
```

	TV	Sales
0	230.1	22.1
1	44.5	10.4
2	17.2	9.3
3	151.5	18.5
4	180.8	12.9

```
In [3]: x = df.iloc[:5, 1].values
y = df.iloc[:5, -1].values
A = (x ** 2).sum()
b = 7.5
B = 2 * (x * (b - y)).sum()
m_opt = - B / (2 * A)
m_opt
```

```
Out[3]: np.float64(0.05558473571748337)
```

```
In [4]: # Calculate minimum
y = pd.DataFrame(df.iloc[:5, -1])
y
```

```
Out[4]:
```

	Sales
0	22.1
1	10.4
2	9.3
3	18.5
4	12.9

```
In [5]: X = pd.DataFrame(df.iloc[:5, 1])
one = [1, 1, 1, 1, 1]
X["one"] = one
X
```

Out[5]: **TV one**

0	230.1	1
1	44.5	1
2	17.2	1
3	151.5	1
4	180.8	1

In [6]: `X_transpose = np.transpose(X)`
`X_transpose`

Out[6]:

	0	1	2	3	4
TV	230.1	44.5	17.2	151.5	180.8
one	1.0	1.0	1.0	1.0	1.0

In [7]: `X_transpose_X = np.dot(X_transpose, X)`
`X_transpose_X`

Out[7]: `array([[1.1086299e+05, 6.2410000e+02],`
`[6.2410000e+02, 5.0000000e+00]])`

In [8]: `X_transpose_y = np.dot(X_transpose, y)`
`X_transpose_y`

Out[8]: `array([[10843.04],`
`[ 73.2 ]])`

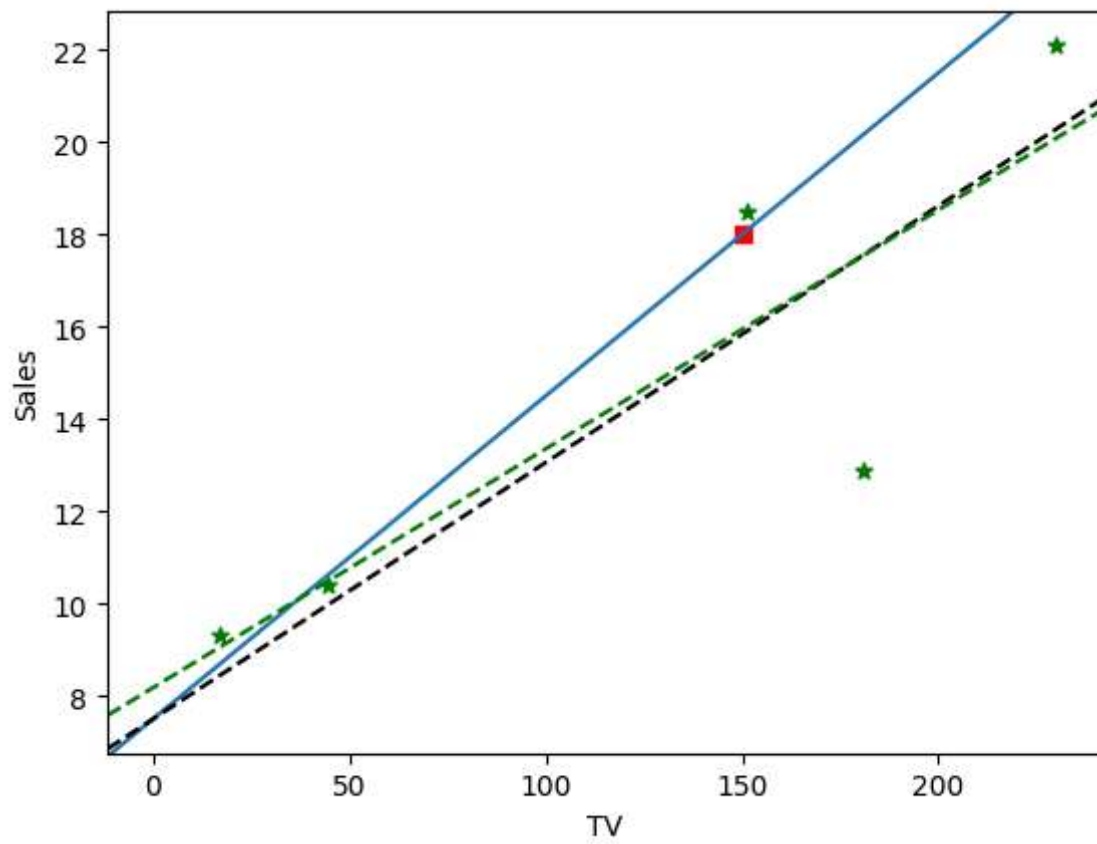
In [9]: `X_transpose_X_inv = np.linalg.inv(X_transpose_X)`

In [10]: `beta = np.dot( X_transpose_X_inv, X_transpose_y )`
`beta`

Out[10]: `array([[0.05176182],`
`[8.17908951]])`

In [11]: `plt.scatter(df.iloc[:5, 1], df.iloc[:5, -1], marker = '*', color = 'g')`
`plt.xlabel(df.columns[1])`
`plt.ylabel(df.columns[-1])`
`plt.axline((150, 18), slope = 0.07)`
`plt.scatter(150,18, marker = 's', color = 'r')`
`plt.axline((0, 7.5), slope = m_opt, color = 'k', linestyle = '--')`

`plt.axline((0, beta.item(1)), slope = beta.item(0), color = 'g', linestyle = '--')`
`plt.show()`



In []: